

• The obvious disadvantage of the map is that it is essentially a trial-and-error procedure which relies on the ability of the human user to recognize certain patterns.

The Tabulation method is a specific step-by-step procedure that is guaranteed to produce a simplified standard-form expression for a function. It is also known as the Quine-McCluskey Method.

References

1. The Problem of Simplifying Truth Functions — W.V. Quine
2. Minimization of Boolean Functions — J.E. McCluskey

• The tabular method of simplification consists of two parts. The first is to find by an exhaustive search all the terms that are candidates for inclusion in the simplified function. These terms are called prime-implicants. The second operation is to choose among the prime implicants those that give an expression with the least number of literals.

3.10 Determination of Prime-implicants

Example 3-13: Simplify the following boolean function by using the tabulation method: $F = \sum(0, 1, 2, 8, 10, 11, 14, 15)$

Table 3-5: Determination of prime-implicants for Example 3-13

(a)				(b)				(c)			
w	x	y	z	w	x	y	z	w	x	y	z
0	0	0	0	0	1	0	0	0, 2, 8, 10	—	0	—
0	0	0	1	0	2	0	0	0, 8, 2, 10	—	0	—
1	0	0	0	0	8	—	0	10, 11, 14, 15	1	—	—
2	0	0	1	0	10	—	0	10, 14, 11, 15	1	—	—
8	1	0	0	2	10	—	0				
10	1	0	1	8	10	1	0				
11	1	0	1	10	11	1	0				
14	1	1	0	10	14	1	—				
15	1	1	1	11	15	1	—				
				14	15	1	1				